

## Highlights

### **Per-window oracle headroom is not learnable: a feedback-delay audit of plug-in selection for time series forecasting**

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- Block filtering flips a plug-in verdict from significant to null.
- Metric and mean-vs-median choices each flip a second verdict.
- Per-window oracle headroom survives per-window arm relabeling.
- Best-arm identity has a median half-life of 5.1% of the forecast horizon.
- The one causally legal axis, lead time, has 0.24% oracle headroom.

# Per-window oracle headroom is not learnable: a feedback-delay audit of plug-in selection for time series forecasting

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## ABSTRACT

Model-agnostic “plug-and-play” modules for time series forecasting—instance normalization, non-stationarity correction, frequency-domain losses—are reported as helpful, yet their gain flips sign across backbones. The natural response is to learn *when* to attach *which* module. In 2,035 single-GPU training runs across four backbones, eight datasets and four horizons, we show that this research program is empty under the only causally legal deployment setting. First, the reported effect is smaller than three near-universal reporting choices: complete-block filtering, the error metric, and mean versus median. Each flips at least one verdict. Second, the per-window oracle that motivates gating is provably invariant to re-labeling which arm produced which error, so its +10.15% headroom is no evidence of learnability; a seed-paired replication nevertheless shows the structure is real, with 79% surviving a seed change. Third, the decisive quantity is the correlation length of best-arm identity: its median half-life is 5.1% of the forecast horizon, whereas deployment forces a one-horizon decision lag. Three selector families accordingly lose to the arm frozen on validation, and the one legal axis, lead time, yields only +0.24%. Only a constant mis-selection is recoverable. We distill the audit into a checklist for adaptive-selection claims.

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## 1. Introduction

The past five years have produced a steady stream of *model-agnostic* additions to deep time series forecasters: reversible instance normalization [1], explicit non-stationarity correction [2], channel clustering [3], frequency-domain training objectives [4]. Each is presented as an orthogonal, architecture-independent, essentially free improvement. Two facts sit uneasily beside that framing.

First, benchmarks have saturated. On ETT / Electricity / Weather, the annual improvement of the best published model has fallen from double digits to a few percent, and unified re-evaluations attribute much of the remaining spread to pipeline details rather than to the method itself [5, 6]; in that regime a reported +0.3% is indistinguishable from a reporting choice.

Second, the sign of the gain is unstable. Practitioners who attach the same plug-in to a linear model and to a patch transformer routinely observe gains of opposite sign. If the gain is conditional, the useful engineering artifact is not another plug-in but a *decision rule*: given (backbone, dataset, horizon), or even given the current window, decide whether to attach a plug-in and which one.

This paper is a systematic, negative, and — we argue — *mechanistic* answer to that decision problem. We do not propose a new plug-in. We build the full conditional-gain matrix, quantify how much of the reported effect is produced by reporting conventions rather than by the method, attack the decision problem with three independent selector families, and finally explain the uniform failure with a single quantitative mechanism: **the correlation length of best-arm identity is one to two orders of magnitude shorter than the feedback delay that deployment imposes.**

Our contributions are fourfold.

1. **Three reporting levers, each larger than the effect.** Complete-block filtering of the main table, the choice of error metric, and the choice between a mean and a median each move the headline by

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more than the effect being claimed, and each flips at least one plug-in verdict (§4). A fourth, reporting a plug-in only at its published default hyper-parameter, accounts for essentially all of one plug-in’s apparent benefit (§5).

2. **Effect size in interpretable units.** We put the block-level plug-in effect on the same scale as the run-to-run noise of a training seed and as the compute it costs (§6): 28% of block-level effects are smaller than same-configuration seed noise, and the arm with the largest compute overhead is also the only one that is significantly *worse* than doing nothing.
3. **An impossibility-style audit.** We prove and empirically verify that the per-window oracle used to motivate gating is invariant to per-window arm re-labeling (§8), then measure the quantity that actually governs learnability — the lagged predictability of best-arm identity — and show it decays to chance well before the legal decision lag (§8.2). A seed-paired replication rules out the trivial explanation that the headroom is training noise (§8.3).
4. **A falsified-hypotheses table, and a closed loop.** Three selector families and two control conditions, all evaluated on identical decision groups against a deployable baseline, are all negative, with a delay sweep pinning the break-even point at  $0.25 \times H$  (§9). Crucially, we also audit the one selection axis that is *not* subject to feedback delay, namely the lead-time index within a forecast (§7); its oracle is worth only +0.24%. Both axes are therefore closed, for two different reasons.

We deliberately report the deployable baseline `best_fixed_val` — the arm chosen on validation data and then frozen — rather than the post-hoc best arm. Much of the apparent success of adaptive selection in the literature disappears against this baseline, and §10 shows that what remains is a validation→test transfer effect, not adaptation.

## 2. Related work and positioning

**Plug-in modules.** Reversible instance normalization [1] removes instance-level distribution shift by denormalizing the output with the input statistics. Non-stationarity correction methods [2] estimate slice-wise statistics and extrapolate them; we implement a *training-free surrogate* (`san_lite`, EWMA extrapolation of slice statistics) so that the arm adds zero trainable parameters and zero per-dataset hyper-parameters, and we label it as a surrogate throughout rather than claiming to reproduce [2]. Frequency-domain losses [4] (`fredf`) add a spectral term with weight  $\alpha$ . Our contribution to this literature is not a new member but an audit of how their gains are reported.

**Benchmark-hygiene work.** A line of work has shown that apparent progress in long-horizon forecasting is sensitive to pipeline details such as `drop_last`, look-back length and checkpoint selection, and that conclusions change once evaluation is unified [5, 7, 8]; a recent reproducible re-evaluation of eight models on fourteen datasets reports that small changes to the experimental setup or to the evaluation metric are enough to overturn claimed state-of-the-art results [6]. We re-run every baseline ourselves under one fixed protocol for exactly this reason and never copy numbers across papers. Section 4 adds new items to that list: which *blocks* enter the aggregate, which *metric* is reported, and which *central estimator* is used are each a significance lever, and we follow [9] on how block-level comparisons should be tested.

**Adaptive / online model selection.** Expert-advice and bandit selection over forecasters is classical [10], and dynamic expert-arbitration for time series is an established family [11], as is drift-triggered adaptation [12]. The regret guarantees, however, assume feedback arrives shortly after the decision; delayed-feedback online learning quantifies the degradation when it does not [13]. The novelty here is empirical and negative: in  $H$ -step-ahead forecasting the feedback delay is *structurally*  $H$  windows, and we measure the identity correlation length that this delay must be compared against. To our knowledge the ratio half-life/ $H$  has not previously been measured for plug-in selection; it is the quantity that decides the problem.

## 3. Protocol

*Grid.* Backbones: DLinear [8], PatchTST [14], TimesNet [15], iTransformer [16] (imported unmodified from a standard library [7]; we do not use its training script). Datasets: ETTh1, ETTh2, ETTm1, ETTm2, Electricity, Exchange, Weather (horizons 96/192/336/720, look-back 96) and ILI (horizons 24/36/48/60,

look-back 36). Arms: **none**, **revin** [1], **san\_lite** (surrogate for [2]), **fredf** [4] ( $\alpha=0.5$ ), plus an  $\alpha$ -sweep pool  $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$  on the plain objective and  $\alpha \in \{0.1, 0.5, 0.9\}$  on the  $\sqrt{H}$ -normalized one (§5).

*Three phases.* Phase 1 is the main gain matrix: 844/844 cells. Phase 2 persists per-window squared errors on validation and test for the nominal four-arm pool: 423/428 cells (5 missing cells are all **TimesNet**  $\times H=720$ ). Phase 3 is the  $\alpha$  sweep over three backbones (**TimesNet** excluded): 768/768 cells, i.e. 96 blocks  $\times$  8 arms (seven  $\alpha$  settings plus a re-trained **none** control). In total we perform 2,035 training runs on a single V100 GPU, in fp32 with AMP enabled on the largest datasets. The three phases overlap by design (2,035 counts runs, not distinct configurations: phase 1 includes 3 seeds on a subset, and phase 2 re-runs the four-arm pool to persist per-window traces).

*Statistical block.* A block is a (dataset  $\times$  horizon  $\times$  backbone) triple, not a dataset. This choice is not cosmetic. With  $k=4$  methods and only the  $N=8$  datasets as blocks — the aggregation level most plug-in papers use — the Nemenyi critical difference is 1.66, wider than the entire observed rank range, so no ranking can be significant at that resolution. At our  $N=128$  blocks it falls to 0.41. We follow [9] for the Friedman/Nemenyi machinery, and all paired tests are Wilcoxon signed-rank reported both raw and with Holm correction inside an explicitly declared family.

*A documented revin pruning.* PatchTST, TimesNet and iTransformer hard-code their own instance normalization, so wrapping them in an external RevIN is mathematically an approximate identity. We therefore pruned that combination *a priori* (skip rule **revin\_on\_internal\_norm\_backbones** in **configs/matrix.yaml**) and kept a verification subset — {ETTh1, Weather}  $\times$  {96, 720} on all three backbones, 12 cells — to check the prediction instead of assuming it. It holds: the median gain over **none** on those 12 cells is +0.00% (Wilcoxon  $p=0.23$ ), and on PatchTST and iTransformer the test MSE is unchanged to four decimals in 7 of 8 cells; the only two cells with a visible effect are TimesNet at  $H=720$  (+0.62% and +5.56%). On DLinear, which has no internal normalization, the same arm has a median gain of +0.98% over 64 cells. No cell of the matrix failed to train — all 844 phase-1 runs report status **ok** — so every **revin** absence below is this design decision, not an outage. The consequence for §4 is mechanical but must be stated: because **revin** exists on 44 blocks by construction, complete-block filtering over four arms can only ever return those 44 blocks.

*Decision group.* For window-level analysis a *decision group* is a (backbone, dataset, horizon) triple for which all arms have aligned per-window error vectors on the same evaluation windows. We have 127 groups for test-side audits. For protocols that must *train* a gate on validation windows we require a provenance marker certifying deterministic loader order (§12); 96 groups qualify. The nominal pool has four arms, but under the pruning above the *realized* arm count per group is 4 in 43 groups, 3 in 83 and 2 in 1 (for the 96 gating groups: 40 and 56). Since oracle headroom grows with the number of arms, we report the realized counts wherever a headroom figure appears and never pool groups of different sizes into a single headroom claim without saying so.

*Baselines.* **none**; **best\_fixed\_val**, the arm with lowest validation MSE, frozen over the test segment (this is the deployable strong baseline and the reference for every gain we report); **best\_fixed\_test**, the post-hoc constant arm (an upper bound on any non-time-varying policy); and the per-window **oracle**.

## 4. How the gain is reported: three significance levers

### 4.1. Lever 1: which blocks enter the aggregate

The standard way to aggregate a plug-in comparison is to keep the blocks in which *all* arms are available and run a paired test on that subset. Absent cells, however, are never absent at random. Whatever prunes a cell — an out-of-budget corner (heavy backbone  $\times$  long horizon  $\times$  large dataset), a crash, or, as here, an explicit design rule — is correlated with the very axes along which the plug-in’s effect varies. In our matrix, 128 blocks exist, but only 44 are complete across four arms, because the **revin** pruning of §3 decides the subset; 32 of those 44 (73%) come from DLinear, the one backbone without internal normalization, which is also the backbone on which frequency-domain losses happen to help most. The filter is therefore a backbone

**Table 1**

Same raw runs, two aggregation conventions. Mean relative gain over **none** (%), Wilcoxon signed-rank  $p$  raw and Holm-adjusted within the family of three arms (each convention is its own family). The significance flip is not an artifact of multiplicity: after Holm correction the complete-block convention still calls **fredf** significant (0.017) while the correct pairwise convention does not (0.76).

| Convention          | Arm             | $N$ | mean          | $p$                | $p_{\text{Holm}}$    |
|---------------------|-----------------|-----|---------------|--------------------|----------------------|
| complete-block (44) | <b>fredf</b>    | 44  | +3.60         | 0.0058             | 0.017                |
| <b>pairwise</b>     | <b>fredf</b>    | 128 | − <b>2.37</b> | 0.44               | <b>0.76</b>          |
| complete-block (44) | <b>san_lite</b> | 44  | −1.29         | 0.056              | 0.11                 |
| <b>pairwise</b>     | <b>san_lite</b> | 128 | − <b>3.94</b> | $5 \times 10^{-8}$ | $1.5 \times 10^{-7}$ |
| complete-block (44) | <b>revin</b>    | 44  | +3.55         | 0.38               | 0.38                 |
| pairwise            | <b>revin</b>    | 44  | +3.55         | 0.38               | 0.76                 |

**Table 2**

**fredf** vs. **none** per backbone, all pairable blocks. The sign flips and the magnitude spans 21 points.  $p_{\text{Holm}}$  is adjusted within the family of four strata; the two significant cells have opposite signs.

| Backbone            | $N$ | mean (%) | $p$                  | $p_{\text{Holm}}$    |
|---------------------|-----|----------|----------------------|----------------------|
| <b>DLinear</b>      | 32  | +4.56    | 0.0036               | 0.011                |
| <b>PatchTST</b>     | 32  | −16.63   | $3.3 \times 10^{-9}$ | $1.3 \times 10^{-8}$ |
| <b>TimesNet</b>     | 32  | +1.41    | 0.15                 | 0.31                 |
| <b>iTransformer</b> | 32  | +1.17    | 0.26                 | 0.31                 |

filter wearing a completeness costume, and the arm it silently re-weights is not the arm that caused the pruning.

Table 1 contrasts the two conventions. Pairwise dropping (each arm compared to **none** on every block where *that pair* exists) is the correct estimator; complete-block filtering is what the literature does.

The direction of the error is not benign. For **fredf** the filter converts a non-significant negative mean into a significant positive one; for **san\_lite** it hides a real, strongly significant degradation. On the complete-block subset a Friedman test rejects ( $\chi^2=30.9$ ,  $p=8.9 \times 10^{-7}$ ,  $\text{CD}=0.71$ ) with mean ranks **fredf** 1.93 < **revin** 2.14 < **none** 2.59 < **san\_lite** 3.34 — a clean-looking ranking produced by a subset that is 73% one backbone.

Table 2 shows why the aggregate is so fragile: conditioned on the backbone, the same plug-in ranges from +4.6% to −16.6%. *Plug-in gains are strongly backbone-conditional*, which is the empirical premise that makes a selection rule attractive in the first place. The rest of the paper tests whether that premise can be converted into a deployable decision.

## 4.2. Lever 2: which error metric is reported

Every number above is MSE, as is standard. Because our evaluation accumulates MAE in the same forward pass, re-running the entire test on MAE costs nothing, and the outcome is not a formality (Table 3). The verdict for **revin** *flips*: it is indistinguishable from **none** on MSE (median +0.03%,  $p_{\text{Holm}}=0.76$ ) and significantly better on MAE (median +1.38%,  $p_{\text{Holm}}=1.4 \times 10^{-5}$ , winning 37 of 44 blocks). The flip is not a ranking artifact: the per-block gains on the two metrics correlate at Spearman  $\rho=0.64$  but agree in sign in only 64% of blocks, i.e. **revin** shifts the error distribution in a way that a squared loss and an absolute loss genuinely disagree about. This is the expected signature of a normalization module: it trims many moderate errors while occasionally enlarging a few large ones, which MAE rewards and MSE does not. For **fredf** and **san\_lite** the two metrics agree ( $\rho=0.97$  and 0.84).

## 4.3. Lever 3: mean or median, and the tail that hides behind it

A third choice is left implicit almost everywhere: papers report a *mean* relative gain and test it with a *rank* test. The two target different functionals, and on this data they disagree about whether a plug-in

**Table 3**

Metric robustness of the plug-in verdicts.

| Arm                                                                             | Metric | $N$ | W/T/L   | Median gain (%) | 95% CI         | $p$                  | $p_{\text{Holm}}$    | $r$  | Verdict       |
|---------------------------------------------------------------------------------|--------|-----|---------|-----------------|----------------|----------------------|----------------------|------|---------------|
| revin                                                                           | MSE    | 44  | 23/0/21 | +0.03           | [-0.52, +0.84] | 0.382                | 0.764                | 0.13 | n.s. (better) |
|                                                                                 | MAE    | 44  | 37/0/7  | +1.38           | [+0.43, +4.05] | $4.8 \times 10^{-6}$ | $1.4 \times 10^{-5}$ | 0.65 | sig. (better) |
| cross-metric: Spearman $\rho = 0.64$ , sign agreement 64%, verdict <b>flips</b> |        |     |         |                 |                |                      |                      |      |               |
| san_lite                                                                        | MSE    | 128 | 30/0/98 | -3.01           | [-4.35, -2.05] | $4.9 \times 10^{-8}$ | $1.5 \times 10^{-7}$ | 0.48 | sig. (worse)  |
|                                                                                 | MAE    | 128 | 44/0/84 | -1.01           | [-1.60, -0.65] | $8.7 \times 10^{-4}$ | 0.002                | 0.29 | sig. (worse)  |
| cross-metric: Spearman $\rho = 0.84$ , sign agreement 84%, verdict consistent   |        |     |         |                 |                |                      |                      |      |               |
| fredf                                                                           | MSE    | 128 | 70/0/58 | +0.45           | [-0.45, +1.27] | 0.438                | 0.764                | 0.07 | n.s. (better) |
|                                                                                 | MAE    | 128 | 79/0/49 | +0.90           | [+0.41, +1.57] | 0.323                | 0.323                | 0.09 | n.s. (better) |
| cross-metric: Spearman $\rho = 0.97$ , sign agreement 93%, verdict consistent   |        |     |         |                 |                |                      |                      |      |               |

Paired Wilcoxon signed-rank per arm against none; blocks are (dataset  $\times$  horizon  $\times$  backbone), matched pairwise so that one arm's pruning cannot shrink another arm's sample. Holm correction is applied within each metric (family size 3). CIs are percentile bootstrap over blocks ( $10^4$  resamples, blocks resampled jointly to preserve pairing). Positive gain = lower error than none.

**Table 4**

The choice of central estimator is a third significance lever.

| Arm      | Metric | Mean (%)     | Median (%)   | Mean-Med | Skew  | < -10% | < -25% | Worst block              | Worst (%) |
|----------|--------|--------------|--------------|----------|-------|--------|--------|--------------------------|-----------|
| revin    | MSE    | +3.55        | +0.03        | +3.52    | +2.19 | 2.3%   | 0.0%   | Exchange/720/DLinear     | -14.1     |
|          | MAE    | +5.01        | +1.38        | +3.63    | +1.57 | 0.0%   | 0.0%   | ETTh1/720/TimesNet       | -5.9      |
| san_lite | MSE    | -3.94        | -3.01        | -0.93    | +0.96 | 21.9%  | 2.3%   | ILI/36/TimesNet          | -30.0     |
|          | MAE    | -1.08        | -1.01        | -0.07    | +1.12 | 7.8%   | 0.0%   | ILI/60/TimesNet          | -19.4     |
| fredf    | MSE    | <b>-2.37</b> | <b>+0.45</b> | -2.82    | -4.47 | 10.9%  | 3.9%   | Electricity/720/PatchTST | -123.4    |
|          | MAE    | <b>-1.18</b> | <b>+0.90</b> | -2.07    | -4.60 | 6.2%   | 3.1%   | Electricity/720/PatchTST | -80.0     |

All columns are computed on the same pairwise-matched blocks as Table 3; only the summary functional changes. Bold rows are sign conflicts, where the mean and the median disagree on whether the plug-in helps at all. "< -10%" is the fraction of blocks whose error is more than 10% worse than none. A paired Wilcoxon test targets the median, so a paper that reports a mean and tests with Wilcoxon is not testing the quantity it reports.

works at all (Table 4). For **fredf** the mean is -2.37% while the median is +0.45%, and the same conflict appears on MAE (-1.18% vs. +0.90%). Neither number is wrong; they answer different questions, and a paper that prints the mean while testing the median is not testing what it reports.

The mechanism is a heavy left tail (skew -4.5), and the tail is not diffuse: it is one identifiable corner of the grid. **fredf** degrades **Electricity** with **PatchTST** by -31%, -61%, -96% and -123% at  $H=96, 192, 336, 720$ , i.e. the failure grows monotonically with the horizon, while the same plug-in helps **DLinear** at  $H=720$  by +19% on **ETTh2** and +25% on **Exchange**. Overall 10.9% of blocks are more than 10% worse than none and 3.9% are more than 25% worse. Reporting a median alone would sell **fredf** as a small free win; reporting a mean alone would sell it as harmful; reporting either alone hides a deployment risk that a practitioner needs to know about. **san\_lite** carries the highest frequency of moderate damage (21.9% of blocks below -10%) but its worst case is milder (-30%).

We therefore report median, mean, a paired bootstrap interval and the tail fractions together for every arm, and we recommend the same in §11. Note that the three levers are independent: they act on which blocks are counted, which loss is measured, and how the counted blocks are summarized, so a paper can be internally consistent and still be off by more than its own effect size on all three at once.

## 5. How the gain is tuned: "plug-and-play" is a tuning effect

Frequency-domain losses introduce one hyper-parameter, the spectral weight  $\alpha$ , and are reported at a fixed default. We sweep  $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$  on the plain objective and  $\alpha \in \{0.1, 0.5, 0.9\}$  on the  $\sqrt{H}$ -normalized one, giving  $96 \times (5 + 3) = 768$   $\alpha$ -cells for analysis. Of these, the 96 plain  $\alpha=0.5$  cells are the phase-1/2 reference and the remaining 672 are trained in phase 3; phase 3's own run count is also 768

**Table 5**

The apparent benefit of a frequency-domain loss is recovered by per-cell tuning of its one hyper-parameter; at the published default the arm is harmful. 96 blocks, three backbones (TimesNet excluded).

| Setting                                                           | mean gain (%) |
|-------------------------------------------------------------------|---------------|
| fixed $\alpha=0.5$ (published default)                            | -3.70         |
| per-cell $\alpha$ tuned on <i>validation</i>                      | +1.79         |
| per-cell $\alpha$ tuned on <i>test</i> (cheating bound)           | +2.29         |
| <b>tuning gap</b>                                                 | <b>+5.48</b>  |
| best <i>single</i> $\alpha$ chosen on validation ( $\alpha=0.1$ ) | +0.12         |
| with $\sqrt{H}$ normalization ( $\alpha=0.1$ )                    | +0.18         |

because it re-trains its own **none** control ( $96 \times (7 \alpha \text{ settings} + 1)$ ), so the two 768s count different things and are not a duplicate. The 96 blocks are 3 backbones  $\times$  8 datasets  $\times$  4 horizons — TimesNet is excluded from the sweep for cost reasons, so this section does not claim to cover it.

Two readings follow from Table 5. (a) The method is not plug-and-play: essentially all of its benefit is recovered by per-cell tuning, and 59.4% of blocks are positive only after tuning. (b) The tuning is genuinely per-cell: no single  $\alpha$  transfers (+0.12%), and the number of distinct optimal  $\alpha$  values that one (backbone, dataset) pair takes across horizons is 1.79. Normalizing  $\alpha$  by  $\sqrt{H}$  reduces that to 1.42 and adds 0.06 points — we report it as an ablation line, not as a contribution.

The gap between validation-tuned (+1.79) and test-tuned (+2.29) is only 0.5 points, so honest tuning is nearly as good as cheating; the problem is not selection noise, it is that the default is wrong for most cells.

## 6. How big the gain is: seed noise and compute cost

Before asking whether a selector can exploit these gains, it is worth asking how large they are in units a practitioner can act on. Two cheap references make the effect size interpretable, and both are computable from artifacts we already have (Table 6).

*Against seed noise.* For 208 configurations we trained three seeds, giving 144 block–arm pairs in which both the arm and **none** have a within-configuration standard deviation. We compare the plug-in effect against  $\sigma_{\text{seed}} = \sqrt{s_{\text{arm}}^2 + s_{\text{none}}^2}$ , the noise scale of a difference of two seed-averaged means; the median  $\sigma_{\text{seed}}$  is 0.0049 MSE. On this scale 28% of block-level plug-in effects are *smaller than the noise of re-running the same configuration with a different seed*, and among the blocks in which the plug-in wins, 27% are in that regime — a different seed could have reversed them. The median effect is  $2.13\times$  the noise scale, so the typical effect is real but not comfortably so, and single-seed per-block claims of the kind that fill plug-in tables are not identifiable. This is a statement about *blocks*, not about the aggregate: the pooled tests of §4 average over 128 blocks and are correspondingly better powered.

*Against compute.* Plug-ins are advertised as free, and in parameter count they are: all three add zero trainable parameters and shift peak memory by less than 0.3 MiB. Wall clock is a different matter. **fredf** costs  $1.03\times$  the training time of **none** for a median +0.45%; **revin** costs  $1.05\times$  for +0.03%; **san\_lite** costs  $1.73\times$  — a 73% training-time surcharge — to be 3.01% *worse*. The arm with by far the largest compute overhead is the only one that is significantly worse than doing nothing on both metrics. Efficiency framing does not rescue any of the three: the best of them buys 0.13 points of median MSE per 1% of extra training time.

## 7. Where conditionality lives: below the block, and the one legal axis

Section 4 established conditionality at the level of the block. Conditionality also exists *inside* a block, and examining it answers a question that the rest of the paper cannot: is there any selection axis that deployment does not forbid?

**Table 6**

Plug-in effects against seed noise and against their compute cost.

| Arm      | $N_{\text{noise}}$ | Within seed noise | Median $ \Delta /\sigma_{\text{seed}}$ | Train-time ratio | Median gain (%) |
|----------|--------------------|-------------------|----------------------------------------|------------------|-----------------|
| revin    | 16                 | 19%               | 3.77                                   | 1.05×            | +0.03           |
| san_lite | 64                 | 27%               | 2.53                                   | 1.73×            | -3.01           |
| fredf    | 64                 | 31%               | 1.74                                   | 1.03×            | +0.45           |

Left half: for the 144 block-arm pairs where both the arm and none have  $\geq 2$  seeds, the noise scale is  $\sigma_{\text{seed}} = \sqrt{s_{\text{arm}}^2 + s_{\text{none}}^2}$  (median  $\sigma_{\text{seed}} = 0.0049$  MSE). “Within seed noise” is the fraction of blocks whose entire plug-in effect is smaller than that scale, i.e. a different seed could flip its sign. Overall 28% of block-level effects and 27% of the wins fall in this regime. Right half: training wall-clock ratio against none in the same block.

**Table 7**

Arm identity is unstable along the forecast horizon, yet the legal axis it opens is empty.

*(a) Does one arm win across all four lead-time quarters?*

| Stratum      | Blocks | Constant argmin | Mean # distinct | Whole-horizon winner suboptimal somewhere |
|--------------|--------|-----------------|-----------------|-------------------------------------------|
| 3-arm pool   | 84     | 65%             | 1.43            | 35%                                       |
| 4-arm pool   | 44     | 39%             | 1.73            | 61%                                       |
| DLinear      | 32     | 38%             | 1.72            | —                                         |
| PatchTST     | 32     | 62%             | 1.44            | —                                         |
| TimesNet     | 32     | 62%             | 1.44            | —                                         |
| iTransformer | 32     | 62%             | 1.53            | —                                         |

*(b) Median relative gain (%) by lead-time quarter*

| Arm      | Q1 (near) | Q2    | Q3    | Q4 (far) |
|----------|-----------|-------|-------|----------|
| revin    | +0.14     | +0.12 | -0.02 | -0.03    |
| san_lite | -5.03     | -3.49 | -2.82 | -1.83    |
| fredf    | -0.36     | +0.32 | +0.93 | +0.48    |

*(c) Headroom of the one causally legal selection axis*

| Reference             | Mean (%)     | Median (%)   | Max (%)      | Blocks with gain                |
|-----------------------|--------------|--------------|--------------|---------------------------------|
| none                  | +3.82        | +1.27        | —            | —                               |
| <b>best fixed arm</b> | <b>+0.24</b> | <b>+0.00</b> | <b>+3.29</b> | <b>56/128 (12% &gt; 0.5 pp)</b> |

$\text{seg}_i$  errors partition the *forecast horizon* into quarters (per-timestep MSE averaged within each quarter), not the test period; the four quarter errors average exactly to the block MSE, which is what makes panel (c) a valid decomposition. Stratifying by arm-pool size is necessary: a 4-arm block has more chances to switch than a 3-arm block, so the two strata must not be pooled. This audit covers all 128 blocks and all four backbones, including TimesNet, which the per-window analysis cannot reach. Panel (c) is the decisive one: unlike the per-window axis, the lead-time index is known at decision time, so switching arms along it needs no feedback and is causally legal — but even its *oracle* buys almost nothing over a single frozen arm.

Our evaluation stores per-timestep test MSE aggregated into four quarters of the forecast horizon ( $\text{seg}_1$  near,  $\text{seg}_4$  far); the four quarter errors average exactly to the block MSE, so they form an exact decomposition. Two features of this audit matter. First, it covers all 128 blocks and all four backbones, including TimesNet, which the per-window analysis of §8–§9 cannot reach (§12). Second, and more important, *the lead-time index is known at decision time*. A policy that applies one arm to the first half of the forecast and another to the second half needs no feedback and incurs no delay. It is the one axis in this paper on which selection is causally legal.

The instability is real (Table 7a). Only 56% of blocks have a single arm that is best across all four quarters, and in 44% of blocks the arm chosen on whole-horizon MSE is beaten by another arm somewhere along the horizon. Stratifying by pool size is necessary — a four-arm block has more opportunities to switch than a three-arm block — and the pattern survives it (65% constant in three-arm blocks, 39% in four-arm blocks). The strata here are 84 and 44 of the 128 *blocks*, which is a different denominator from the 83/43/1 *decision groups* of §3: this audit reads block-level segment errors and therefore covers TimesNet and all 128

**Table 8**

Lagged agreement of best-arm identity, 127 groups. Chance =  $\sum_k p_k^2 = 0.395$ ; after per-window re-labeling the measured agreement is 0.307, matching chance. Only  $\ell = H$  is causally legal.

| Lag $\ell$       | agreement    | excess        | groups $z > 2$ |
|------------------|--------------|---------------|----------------|
| 1 window         | 0.864        | +0.469        | 127/127        |
| $H/4$            | 0.467        | +0.072        | 93/127         |
| $H/2$            | 0.415        | +0.020        | 72/127         |
| <b>H (legal)</b> | <b>0.389</b> | <b>−0.007</b> | <b>37/127</b>  |
| $2H$             | 0.397        | +0.002        | 41/127         |

blocks, whereas a decision group additionally requires aligned per-window traces and there are 127 of those. The profiles are also interpretable (Table 7b): **fredf** is mildly harmful at short lead times (median  $-0.36\%$  in Q1) and helpful further out ( $+0.93\%$  in Q3), consistent with a spectral term that trades local fidelity for global shape, whereas **san\_lite**’s damage is worst near the origin ( $-5.03\%$  in Q1) and decays with lead time.

And yet the axis is empty (Table 7c). An *oracle* that picks the best arm independently for each quarter beats **none** by  $+3.82\%$  on average, which sounds like a target — but almost all of that is the value of choosing an arm at all, not of switching. Measured against the best single frozen arm in the same block, the per-lead-time oracle is worth  $+0.24\%$  on average, exactly  $0.00\%$  at the median, and above half a point in only 12% of blocks; its ceiling anywhere in the grid is  $+3.29\%$ . Since this is an oracle, no deployable per-lead-time rule can do better.

The two axes are thus closed for opposite reasons, and together they close the loop: the per-window axis has large headroom that expires before it can be acted on (§8–§8.2), while the lead-time axis can be acted on freely but has no headroom to act on. We note that measuring the *deployable* version of the lead-time policy would require per-lead-time errors on the validation split, which our pipeline does not persist; given a  $+0.24\%$  oracle ceiling we did not add them, but persisting them is a one-line change that we recommend to anyone who wants to close this axis empirically rather than by ceiling argument.

## 8. Why per-window selection cannot work

### 8.1. The per-window oracle is label-invariant

Every gating paper motivates itself with a per-window oracle: if for each evaluation window  $t$  we could pick the arm  $k$  with the smallest error  $e[k, t]$ , we would gain 10%. Our matrix reproduces this: per-window oracle vs. **none** is  $+10.15\%$  over 127 groups (and  $+9.74\%$  vs. **none** on the 96 gate-eligible groups, where the deployable **best\_fixed\_val** gains only  $+2.12\%$ ).

This statistic cannot support the inference that is routinely drawn from it.

**Proposition 1 (Re-labeling invariance).** *Let  $e[\cdot, t] \in \mathbb{R}^K$  be the arm errors in window  $t$  and let  $\sigma_t$  be a permutation of  $\{1..K\}$  drawn independently for each  $t$ . Then  $\frac{1}{T} \sum_t \min_k e[\sigma_t(k), t] = \frac{1}{T} \sum_t \min_k e[k, t]$ .*

PROOF.  $\min_k e[\sigma_t(k), t]$  depends only on the multiset  $\{e[1, t], \dots, e[K, t]\}$ , which  $\sigma_t$  leaves unchanged.

The oracle therefore measures only the *dispersion* of arm errors within windows, not any association between window and best arm. Independently permuting arm labels per window destroys every learnable pattern and leaves the headroom bit-identical — we verified this numerically in all 127 groups (127/127 invariant). Any non-degenerate arm pool has a large per-window oracle; “oracle  $\gg$  best fixed” is a property of variance, not of learnability.

The quantity that does govern learnability is the lagged predictability of the best-arm identity  $a_t = \arg \min_k e[k, t]$ . We measure the agreement  $\Pr[a_t = a_{t-\ell}]$  against the chance level  $\sum_k p_k^2$ , where  $p_k$  is the marginal win frequency of arm  $k$ .

Table 8 is the core of the paper. At lag 1 the identity is enormously predictable ( $+0.469$  over chance, every single group significant). At the only lag a deployed system may use it is *at chance* ( $-0.007$ ). The signal exists; it expires before it can be acted on.

**Table 9**

Excess-over-chance agreement of best-arm identity as a function of absolute lag, pooled over the 127 decision groups.

| lag    | 1    | 4    | 16   | 32   | 64   | 128   | 512  |
|--------|------|------|------|------|------|-------|------|
| excess | .469 | .321 | .163 | .092 | .041 | −.007 | .005 |

**Table 10**

The decay is not an averaging artifact: three independent stratifications (backbone, dataset, horizon) give 20 usable strata and no exception.

| Criterion                              | Result               |
|----------------------------------------|----------------------|
| strata with half-life $< 25\%H$        | <b>20/20</b>         |
| strata with excess $> 0.05$ at lag $H$ | <b>0/20</b>          |
| most signal-like stratum               | Exchange, $+0.032$   |
| longest relative correlation           | $H=24$ , $19.2\%$    |
| range of median half-life              | $3.0 - 15.3$ windows |

The per-group  $z$  column is a dispersion diagnostic, not an inference. The  $z > 2$  counts use the i.i.d. binomial reference  $z = (\widehat{\text{Pr}} - \text{chance}) / \sqrt{\text{chance}(1 - \text{chance}) / (n - \ell)}$ , which is anti-conservative here: windows overlap at stride 1, so the  $n - \ell$  agreement indicators are far from independent and  $|z|$  is inflated by roughly the number of windows per independent block. We keep the column because its *shape* is the point. At the legal lag the counts are exactly symmetric — 37/127 groups with  $z > 2$  and 37/127 with  $z < -2$ , and 57/127 with positive excess — which is the signature of an over-dispersed null rather than of a weak but real signal. Dividing each  $z$  by the crude block factor  $\sqrt{H}$  leaves 0/127 groups significant in either direction. By contrast, at lag 1 the 127/127 count is one-sided and survives any such deflation. No claim in this paper rests on these  $z$  values; the decisive quantities are the mean excess ( $-0.007$ ) and the half-life of §8.2.

*Why the legal lag is  $H$ .* Window  $t$  predicts the interval  $[t + L, t + L + H)$ . Its realized error is observable only after that interval closes, i.e.  $H$  windows later. A selector that decides for window  $t$  using any feedback from window  $t' > t - H$  is using information that does not exist at decision time. This is not a modeling choice but an arithmetic property of  $H$ -step forecasting, and it is the reason the entire literature’s zero-delay ablations look so much better than anything deployable.

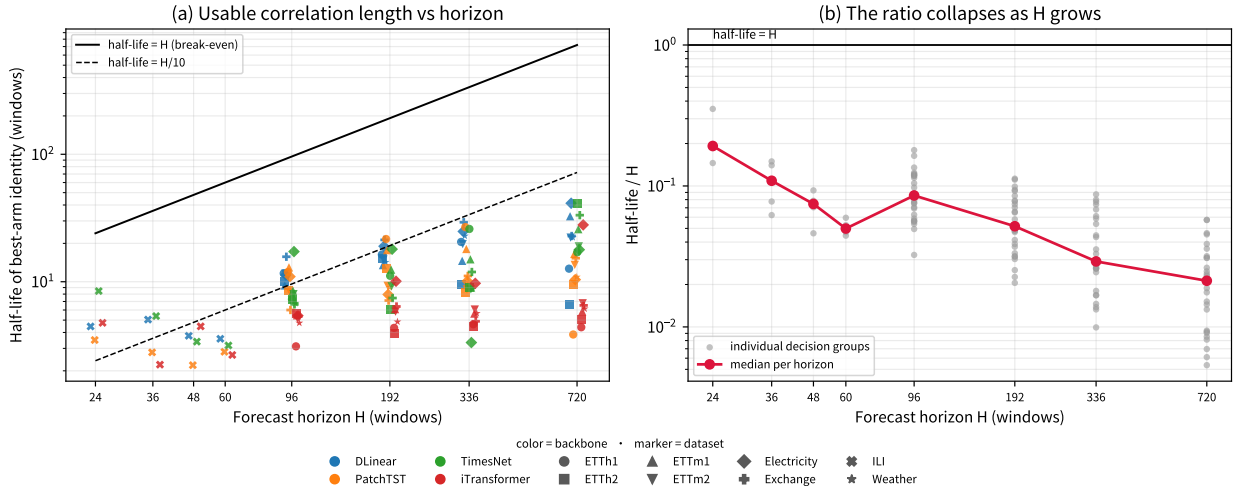
## 8.2. How long does the signal live?

We summarize the decay of the excess agreement by its *half-life*: the absolute lag (in windows) at which excess-over-chance falls to half of its lag-1 value.

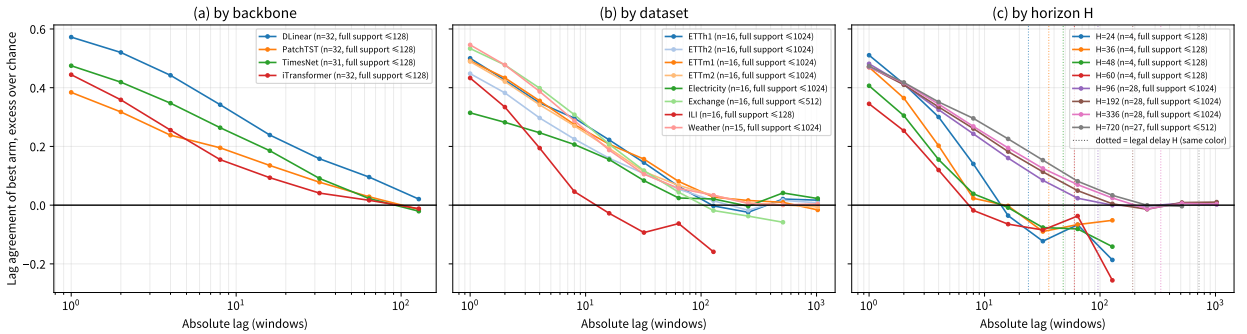
The median half-life is **9.0** windows (mean 11.3). Because  $H$  varies across groups we also report the ratio directly: half-life/ $H$  has mean **6.1%** and median **5.1%**. Across 127 groups, 86% have a half-life below  $H/10$ . The modal arm already wins 48.8% of windows (normalized win-distribution entropy 0.88), so a majority-vote constant policy captures most of the achievable identity mass.

*Is this an average artifact?* A reviewer will ask whether some backbone, dataset, or short horizon does carry usable signal that the average hides. We recompute the audit under three independent stratifications (backbone, dataset, horizon), 20 usable strata, none under-powered.

*Horizon scaling is the wrong way round.* The absolute half-life does grow with  $H$  (Spearman  $+0.74$  over the eight horizon-level medians,  $+0.49$  over all 127 groups; 4.6 windows at  $H=24$ , 15.3 at  $H=720$ ) but far more slowly:  $H$  grows  $30\times$  while the half-life grows  $3.3\times$ . Hence half-life/ $H$  *decreases* with  $H$  (Spearman  $-0.88$  over the horizon-level medians,  $-0.73$  over all groups): 0.192 at  $H=24$  down to 0.021 at  $H=720$ . Long-horizon forecasting — precisely where plug-ins are marketed — is where window-level plug-in selection is most hopeless. Figures 1 and 2 plot the two views.



**Figure 1: The usable correlation length never reaches the decision lag.** (a) Half-life of best-arm identity vs. forecast horizon, one marker per decision group (127; color = backbone, marker = dataset), log-log. The solid line is the break-even condition  $\text{half-life} = H$ ; no group comes within an order of magnitude of it, and most sit below the  $H/10$  dashed line. (b) The ratio  $\text{half-life}/H$  per horizon: gray dots are groups, red is the median, falling from 0.19 at  $H=24$  to 0.021 at  $H=720$ . The decline is monotone *within* each of the two horizon families (24/36/48/60 for ILI, 96/192/336/720 for the rest); the small step up at  $H=96$  is the change of family, not a reversal.



**Figure 2: The same decay under three independent stratifications.** Excess-over-chance agreement of best-arm identity vs. absolute lag (windows), by backbone (a), dataset (b) and horizon (c). Only lags with full support across the groups in a stratum are drawn. Every curve reaches chance by lag  $\approx 10^2$ ; in (c) the dotted vertical lines mark each stratum's legal decision lag  $H$  (same color), and every curve has already flattened at or before its own line. No stratum is an exception.

### 8.3. The headroom is real, just late

Proposition 1 is constructive; a skeptic may answer that real labels are not random permutations. We therefore give a purely empirical second leg. For 48 cells we retrain with only the training seed changed (2021  $\rightarrow$  2022) and ask how much of the per-window best-arm identity survives.

The key statistic is the *cross-seed oracle*: choose arms with seed A's per-window arg min, settle the bill on seed B's errors. It still uses future information, but only the part of the arm identity that is reproducible, so it upper-bounds any selector that relies on reproducible structure — including a static gate given perfect features.

The last three rows of Table 11 are a necessary caveat: if a backbone converges to the same solution under both seeds, reproducibility is trivial and the audit has no discriminative power for it (near-convex models such as DLinear are the worst case). With same-arm error correlation 0.989 and block-level MSE moving by 1.33%, the model does change, yet *which window wants which plug-in* stays stable. So the finding is not “there is nothing to learn” but the sharper “the thing to learn is real, and it is unavailable in time”.

**Table 11**

A seed-paired replication shows the per-window structure is real and largely reproducible, so the failure of every selector cannot be explained as “there is nothing there to learn”.

| Quantity                                      | Value                |
|-----------------------------------------------|----------------------|
| paired cells / seeds                          | 48 / {2021, 2022}    |
| same-seed oracle vs. best fixed               | +7.26%               |
| <b>cross-seed oracle (reproducible bound)</b> | <b>+5.74%</b>        |
| reproducible share of headroom                | 79%                  |
| cross-seed identity agreement                 | 0.788 (chance 0.374) |
| pairs with $z > 2$                            | 48/48                |
| pairs where cross-seed oracle still wins      | 46/48                |
| same-arm per-window error correlation         | 0.989                |
| block MSE fluctuation across seeds            | 1.33%                |
| best fixed arm unchanged across seeds         | 88%                  |

**Table 12**

Window-level gating. Gains in % relative to the stated reference; bf = best\_fixed\_val. “in-pool” trains and applies within each group; “LODO” holds out a whole dataset; “learnable” restricts to groups with  $\geq 1000$  validation windows.

| Protocol  | head | $N$ | vs none | vs bf | oracle |
|-----------|------|-----|---------|-------|--------|
| in-pool   | reg  | 96  | +1.50   | −0.67 | +9.74  |
| in-pool   | clf  | 96  | +1.49   | −0.65 | +9.74  |
| learnable | reg  | 72  | +2.46   | −0.66 | +9.10  |
| LODO      | reg  | 76  | +1.23   | −1.56 | +10.46 |

## 9. Three selector families, one verdict

We now test the decision problem directly. All numbers are relative to the deployable best\_fixed\_val baseline on identical decision groups.

### 9.1. Static complexity-feature gating

Per window we compute 24 cheap statistics in five groups — non-stationarity (**w\_ns**), trend and autocorrelation (**w\_sd**), cross-channel correlation (**w\_ch**), spectral concentration (**w\_fr**) and predictability/entropy (**w\_pc**) — all  $O(n \log n)$  from the raw look-back window, with no access to the target. A gradient-boosted tree is trained on validation windows (either regressing per-arm gain and thresholding, or classifying the arm directly) and applied to test windows.

Gating loses to a frozen arm by 0.67 points (positive in 22% of groups, Wilcoxon signed-rank across groups  $p=3.1 \times 10^{-6}$ ), captures a *negative* fraction of the oracle headroom, and its per-window accuracy (0.416) is below the majority-class rate (0.486). Restricting to data-rich groups does not help (−0.66); requiring cross-dataset transfer makes it worse (−1.56). Stratifying by the number of validation windows ( $< 200$ ,  $200-1000$ ,  $1000-5000$ ,  $\geq 5000$ ) gives −0.46, −1.09, −0.60, −0.73: the failure is not a sample-size artifact. The LODO row covers 76 of the 96 gating groups: leaving out one dataset requires at least three datasets sharing a (backbone, horizon, arm-pool) shape, which excludes ILI entirely (each ILI horizon exists on one dataset only) and excludes the four-arm PatchTST/iTransformer shapes at  $H=96$  and  $H=720$ , which span two datasets. The exclusions are structural, not outcome dependent.

*Falsification controls.* Two controls separate “window-level information” from “switching per se”: a random switcher matched to the gate’s switching frequency and arm marginals (−1.04 vs bf), and the same model trained on column-shuffled features, i.e. a same-capacity noise-fitting baseline (−0.54). The gate’s net advantage is +0.36 over random (66% of groups) and −0.22 over shuffled (26%). The honest reading: after the overfitting baseline is subtracted, the contribution of window-level features is not distinguishable from

**Table 13**

Same-distribution split diagnostic: gate fitted and evaluated inside one split (chronological 70/30), removing both label optimism and val→test shift.  $\text{bf}_{\text{split}}$  is the fixed arm chosen on the fitting segment. Groups with  $< 200$  fitting or  $< 100$  evaluation windows are skipped.

| fit/eval split | $N$ | vs none | vs $\text{bf}_{\text{split}}$ | win | oracle |
|----------------|-----|---------|-------------------------------|-----|--------|
| val internal   | 81  | +2.01   | −1.18                         | 35% | +11.35 |
| test internal  | 111 | +2.33   | − <b>0.53</b>                 | 38% | +10.37 |

**Table 14**

Online selection with legal  $H$ -window delay (123 groups). Pool sizes are nominal; realized arm counts per group are given in §3. “delay cost” is the no-delay advantage.

| Pool (nominal) | vs none | vs bf         | no-delay | delay cost |
|----------------|---------|---------------|----------|------------|
| $\leq 4$ arms  | +1.19   | − <b>0.87</b> | +2.47    | 3.34 pp    |
| $\leq 11$ arms | +1.31   | −0.82         | +2.84    | 3.66 pp    |

zero, and we do not claim it. Leave-one-group-out feature ablation is consistent — removing any of the five feature groups changes the gain by at most 0.13 points, and removing the entropy group even *improves* it.

*Same-distribution split diagnostic.* The gate above is trained on validation windows and applied to test windows, so two confounds remain: validation errors are mildly optimistic (that split also drives early stopping) and there is a val→test shift. We remove both by re-running the identical gate *inside a single split*: the first 70% of windows chronologically to fit, the last 30% to evaluate, with the fixed-arm baseline  $\text{bf}_{\text{split}}$  also chosen on the fitting segment, so neither side sees the evaluation segment. With **source=val** the optimism is present on both sides and cancels; with **source=test** the labels have *never* been touched by early stopping or checkpoint selection, so the label bias is exactly zero and all backbones are covered. Both are diagnostic upper bounds, not deployable policies.

The verdict is unchanged (Table 13). Under zero drift and zero label bias the gate still loses to a frozen arm chosen from the same information (−0.53, positive in 38% of groups,  $p=0.19$ ), and its per-window accuracy (0.411) is still below the majority-arm rate (0.486), beating it in only 13% of groups. Its share of the +10.4% oracle headroom is small and, more importantly, unstably estimated: as a ratio of aggregate means it is  $+2.33 / +10.37 = 22\%$ , whereas the per-group ratio has median 4.6% and mean 3.3% with an interquartile range of −9% to +35% and a minimum of −579%, because groups whose oracle headroom is nearly zero put a vanishing denominator under a noisy numerator. We report all three rather than pick one, and we do not rest any conclusion on them: the load-bearing fact is the comparison against the frozen arm, which needs no ratio. This rules out “the validation split was used for early stopping” and “val→test shift” as explanations: what is missing is window-level signal, not clean labels. One further detail sharpens the main result: in the test-internal setting the gate is on par with the *validation*-chosen arm (+0.08) while losing to the split-chosen arm (−0.53), and the split-chosen arm coincides with the evaluation-segment argmin in 70% of groups versus 47% in the val-internal setting. The small gains a window gate appears to make over **best\_fixed\_val** elsewhere are arm-identity mismatch between splits, not per-window adaptation.

## 9.2. Delayed online error feedback

Static features must transfer across a distribution shift; deployment has a stronger signal available, namely the *realized* per-arm error on past windows. We run follow-the-leader over a sliding feedback window of 200, with the delay set to exactly  $H$  windows, and additionally a no-delay cheating variant. This is the delayed-feedback regime of [13], with the distinguishing feature that the delay is not an implementation nuisance but is fixed by the forecast horizon itself.

The legal configuration loses (−0.87, positive in 20% of groups) while the illegal one wins by +2.47: a 3.34-point delay cost. This is the same conclusion as Table 8, reached with a completely different estimator and no learned model at all. The count is 123 of the 127 decision groups; the four excluded ones are

**Table 15**

Delay  $\times$  feedback-window sweep, gains vs. bf (%). “fb” is the sliding feedback-window length. Only  $\ell \geq 1H$  is causally legal.

| delay     | fb= 50 | 200   | 1000  | 3000  | legal?     |
|-----------|--------|-------|-------|-------|------------|
| 0         | +4.59  | +2.19 | +0.96 | +0.88 | cheat      |
| 0.25H     | +0.83  | +0.07 | +0.22 | +0.43 | cheat      |
| 0.5H      | -0.49  | -0.64 | +0.01 | +0.32 | cheat      |
| <b>1H</b> | -1.36  | -0.77 | -0.13 | +0.12 | <b>yes</b> |
| 2H        | -0.92  | -0.61 | -0.01 | +0.22 | yes        |

**Table 16**

Decomposition of the online policy’s gain over none. The time-varying component is negative; the whole effect is the correction of a constant mis-selection.

| Component                              | Value  |
|----------------------------------------|--------|
| bf_test – bf_val                       | +1.36% |
| (groups where val picks the wrong arm) | 34%    |
| online – bf_test                       | -2.32% |
| (groups where this is positive)        | 5%     |

Exchange at  $H=720$  on all four backbones, where the legal delay (720 windows) exceeds the number of evaluation windows (399), so no legal decision can be made at all. That exclusion is itself an instance of the thesis, not a limitation of the estimator.

*Delay sweep.* Sweeping the delay in multiples of  $H$  (all cells evaluated on one common test segment, skipping the warm-up of the longest delay — omitting this dilutes long-delay rows toward zero and produces the spurious conclusion that  $2H$  beats  $1H$ ) gives a clean break-even point (Table 15).

Averaged over feedback windows on the 103 groups with common support, the break-even delay is  $0.25 \times H$ : at  $1 \times H$  the policy is  $-0.38\%$ , at zero delay  $+1.89\%$ . Note also the interaction: short feedback windows are best when the delay is small and worst when it is legal, because adaptation speed is only useful if the information is fresh.

### 9.3. Horizon-aware checkpoint selection

The third family is zero-compute: instead of selecting the checkpoint by overall validation MSE, we select it by a horizon-aware read of the same training logs (worst forecast segment, long-segment MSE, slope-penalized, rank aggregation, last epoch). Over all 844 cells, none beats the validation-MSE baseline (gains  $-0.002$  to  $-0.074$ ; taking the last epoch is  $-2.03$ ,  $p=6.8 \times 10^{-13}$ ), while the post-hoc oracle checkpoint would give  $+0.92$ . Even re-reading existing logs yields nothing.

## 10. What is actually recoverable

If online feedback loses to **best\_fixed\_val**, what did its  $+1.19\%$  over **none** consist of? Decomposing against the post-hoc constant arm **best\_fixed\_test** settles it (Table 16).

The time-varying component is negative. The only real effect is that validation-based arm choice is wrong in a third of groups, and delayed feedback partially corrects that *constant* choice. This is a legitimate, modest, deployable claim: *use online feedback to fix plug-in mis-selection, not to adapt per window*. Any paper claiming window-level adaptation must publish the **best\_fixed\_test** column, or the claim is unfalsifiable.

## 11. A checklist for adaptive-selection claims

The audit generalizes beyond plug-ins to any “learn which model/module to use when” claim. We propose five tests, all cheap once per-window errors are persisted:

1. **Re-labeling test.** Recompute your oracle after permuting arm labels independently per window. If it does not move (Prop. 1), it is not evidence of learnability and must not be cited as headroom.
2. **Correlation-length test.** Report the half-life of best-arm identity in units of the decision lag. If half-life/lag  $\ll 1$  (here: mean 0.061, median 0.051), no causal selector can work; ours is  $< 0.25$  in 20/20 strata.
3. **Deployable-baseline test.** Compare against the arm frozen from validation, and publish the post-hoc constant arm too, so that “adaptation” is separated from “fixing a bad constant choice” (§10).
4. **Shuffled-feature test.** Train the identical model on column-shuffled features. The difference, not the raw gain, is your evidence (here:  $-0.22$  points).
5. **Same-split test.** Before blaming distribution shift or noisy labels, fit and evaluate the selector inside one split, chronologically, with the fixed-arm baseline chosen on the same fitting segment. A selector that cannot win with zero drift and unbiased labels (Table 13) will not win in deployment, and this costs no additional training.

Tests 1–2 say whether the target is reachable, test 5 whether the pipeline is to blame, tests 3–4 whether a reported win is a win.

For the reporting side, §4–§6 add four more that cost nothing but a few lines of analysis code: aggregate with pairwise dropping and publish the per-arm block counts; report every verdict on at least two error metrics, since one of our three arms changes verdict between MSE and MAE; report the median, the mean and the tail fractions together, because the first two disagree in sign for one of our arms and the tail is where the deployment risk lives; and put the effect next to the run-to-run noise of a training seed and next to the wall-clock time it costs, because 28% of our block-level effects are inside seed noise and the arm with the largest compute surcharge is the one that hurts most. Finally, report any plug-in both at its published default and per-cell tuned, with the gap stated (§5).

## 12. Threats to validity

*A provenance bug we found in ourselves.* An earlier version of this study trained the window gate on validation error traces whose order had been permuted by a shuffling data loader (the library disables shuffling for `test` but not for `val`). Features of window  $i$  were paired with the error of a random window  $j$ . Every gating/control/ablation number was therefore invalid, while all test-side audits were unaffected (the test loader never shuffles) and all per-arm *means* were unaffected (a mean is permutation-invariant, so `best_fixed_val` survived). We re-ran 328 cells, added a machine-checked provenance marker asserting deterministic loader order, and made the analysis code refuse unmarked arrays rather than silently reuse them. We report this because the corrected gating number ( $-0.67$ ) is smaller in magnitude than the buggy one ( $-1.76$ ): the bug was making our negative result look stronger than it is.

*Coverage asymmetry.* The test-side audits (Tables 8, 14) use 127 decision groups; only 96 have a certified deterministic validation window order and can therefore train a gate (Table 12). Table 17 makes the gap explicit rather than asserting it is harmless. All 31 missing groups are `TimesNet`, for a bookkeeping reason: those validation traces predate the provenance marker above. The question is whether they differ on the quantities that decide learnability. They do not: half-life over the legal delay is 0.053 versus 0.049 for covered groups, and excess above chance at delay  $H$  is  $-0.002$  versus  $-0.008$  — the same distribution, both far from the threshold at which per-window selection could pay. The missing groups have a *larger* oracle headroom ( $+11.4$  vs  $+9.7$ ), which is precisely the direction that would make gating look good if headroom were the operative variable; it is not. Filling the gap would add samples, not reverse the sign. Two independent audits also cover all four backbones including `TimesNet` and reach the same verdict: the test-internal split diagnostic (Table 13,  $-0.53$  there) and the lead-time audit of §7, which uses all 128 blocks.

*Other caveats.* (i) A small number of windows produce NaN features when a channel is constant inside the window; the tree model handles missing values natively and we do not impute. (ii) `san_lite` is a training-free surrogate, not the original method, and the compute surcharge we report in §6 is that of the surrogate. (iii) The  $\alpha=0.5$  reference comes from phase 1/2, not the sweep; we aligned seeds and checked the reproducibility

**Table 17**

Explicit coverage gap between the test-side audits and the gating protocols, with the audit statistics that decide learnability.  $t_{1/2}/H$  is the median arm-identity half-life divided by the legal feedback delay;  $\text{exc}@H$  is the mean excess above chance at delay  $H$ .

| backbone     | audit | gate | oracle | $t_{1/2}/H$ | $\text{exc}@H$ |
|--------------|-------|------|--------|-------------|----------------|
| DLinear      | 32    | 32   | +14.73 | 0.076       | -0.010         |
| PatchTST     | 32    | 32   | +4.66  | 0.047       | -0.010         |
| iTransformer | 32    | 32   | +9.83  | 0.030       | -0.005         |
| TimesNet     | 31    | 0    | +11.40 | 0.053       | -0.002         |
| covered      | 96    | 96   | +9.74  | 0.049       | -0.008         |
| missing      | 31    | 0    | +11.40 | 0.053       | -0.002         |

of duplicate cells. (iv) Missing cells concentrate in the expensive corner, so every cross-backbone average in this paper is computed on a single common block set. (v) The lead-time audit of §7 resolves the horizon into quarters, not individual steps; a finer partition could expose switching that quarters average away, though it cannot raise the oracle ceiling by much, since the quarter-level oracle is already within 0.24% of a frozen arm. (vi) **revin** was deliberately pruned outside **DLinear** (§3), so its metric-level verdict in Table 3 rests on 44 blocks that are predominantly one backbone, and we do not generalize it; the 12-cell verification subset supports the no-op prediction but is too small to support a metric-level claim. (vii) All conclusions are for the four backbones and eight datasets studied at look-back 96 (36 for ILI); zero-shot foundation models are out of scope (they require bf16, unavailable on the target hardware, and would introduce an incomparable baseline).

### 13. Conclusions

Plug-in gains for time series forecasting are strongly conditional — the same frequency-domain loss moves a linear backbone by +4.6% and a patch transformer by -16.6% — and the way the community aggregates, measures, summarizes and tunes them inflates the reported effect past the effect itself: block filtering, the metric, the choice of mean or median, and the default hyper-parameter each move the headline by more than the claim. That conditionality invites a selector, and a per-window oracle appears to promise 10% for building one. We showed that this promise is a statistical mirage (the oracle is invariant to per-window arm re-labeling), that the underlying structure is nonetheless real and reproducible across training seeds, and that it is unusable anyway because the identity of the best arm decorrelates with a median half-life of 5.1% of the forecast horizon (mean 6.1%) while deployment forces a decision lag of exactly one horizon. Three selector families fail accordingly and the break-even delay is a quarter of a horizon. The one selection axis that deployment does *not* forbid, the lead-time index, turns out to have an oracle headroom of +0.24%, so it is closed for the opposite reason. The only recoverable gain is the correction of a constant mis-selection. We recommend that the field stop tuning gates and start publishing the audits above.

### CRediT authorship contribution statement

**Mianhan Liu:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Chen Chen:** Validation, Investigation, Writing – review & editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model to assist with copy-editing, LaTeX formatting and the drafting of analysis and plotting scripts. All experimental design, statistical decisions, numerical results and scientific conclusions were produced and verified by the authors, who take full responsibility for the content of the publication.

## Relation to other work by the authors

This study and the companion manuscript [17] share an experimental infrastructure and a research programme but address disjoint questions and report disjoint results: this paper asks whether per-window plug-in selection is learnable under a legal feedback delay, and the companion asks whether a stored per-sample artifact is in the order its consumer assumes. Neither manuscript's claims depend on the other's; the overlap is disclosed here so that editors and reviewers can assess it directly.

## Data availability

All datasets used in this study are public standard benchmarks (ETT, Electricity, Exchange, Weather, ILI) and are not redistributed here. The artifacts required to recompute every number in this paper are released at <https://github.com/oscarliu2019/plugin-selection-audit>: the experiment configurations; per-run result tables for the 844 phase-1 and 423 phase-2 runs; per-block aggregates for the 768 phase-3 runs of the  $\alpha$  sweep; the window-level feature table; the per-group outputs of every audit reported here (decay curves and half-lives, oracle re-labeling, delayed-feedback and lead-time sweeps, seed replication, split diagnostics, coverage gap); and the audit, statistics, table and figure generators. The raw per-window error arrays written during training are not redistributed; they are regenerated by re-running the phase-2 configuration with the `--save-window-mse` flag. Each table states the artifact it is regenerated from, and the commands are listed in the repository README. A verification script, `tools/verify_paper_numbers.py`, re-derives every quantitative claim in this manuscript from those artifacts and fails on any mismatch; it prints the claim identifier, the section, the value printed here and the value recomputed.

The repository is also archived permanently in Software Heritage: the revision from which this manuscript was produced is `swh:1:rev:ec90998cd56de249ab48eb381934a93a82d3e12a` (origin <https://github.com/oscarliu2019/plugin-selection-audit>, visit `swh:1:snp:cd0b09f87d11cd06b53af22f6b0439fd8687a009`).

The exact release from which every number in this manuscript was computed is archived, with all artifacts, at <https://doi.org/10.5281/zenodo.22809153> (DOI: 10.5281/zenodo.22809153); the repository above is its development mirror.

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